

# Homework 2: Numeric Summaries

Name:

Class Start Time:

For the homework assignment we will be discussing Lego's! It's a data set that one of my mentors actually collected as her and another prof. were outraged by the cost of legos and wanted to figure out when there was a good deal. The data comes from three distinct types of "Themes". One is City theme'd, one is Friends theme'd and one is Duplo theme'd. The first two are more traditional style lego sets while the third is marketed to toddlers with extra large block size (to minimize choking hazards). Please feel free to google these to have a better idea about these themes.

## Question 1

1. What is the observational unit in this study? What was the physical thing that we recorded information (price, number of pieces) on?

The observational unit is a single lego set

2. Please identify if the three variables listed are numeric or categorical. If they are numeric indicate if they are discrete or continuous. If they are categorical, please mention if they are nominal or ordinal. You do not need to justify your answer unless you are unsure.

Lego Theme  
Categorical, nominal

Amazon Price  
Numeric, continuous

Number of Pieces  
Numeric, discrete

3. Which variable is our response variable (i.e. the dependent variable)?

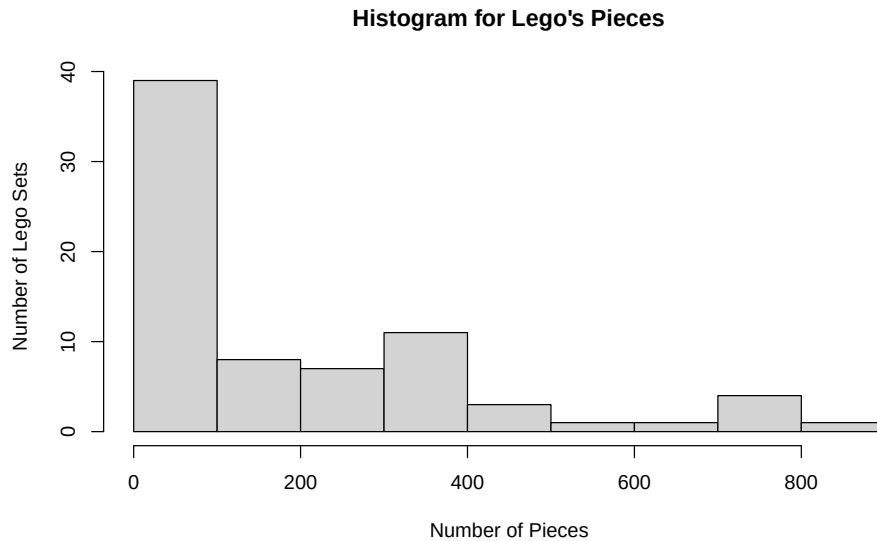
The price on amazon of a lego set

4. Which of the three variables would create easily identifiable subpopulations? What would the subpopulations be?

Theme creates the most obvious subpopulations (Duplo, Friends, City). Other answers are possible but of less obvious nature.

## Question 2

1. Below is the histogram of the number of pieces in a lego set. On the side please indicate if it's symmetric or skewed; if there are outliers; and how many modes it has.



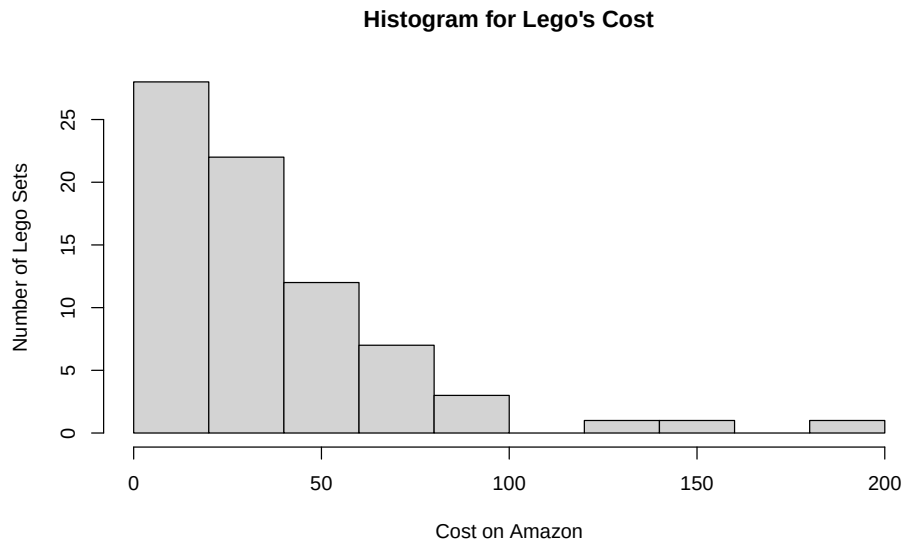
Very skewed right, no outliers, and unimodal(ignore the two minor upticks on the right side of the graphs and focus on the big picture)

2. Would you expect the mean or median to be larger? Why?  
Mean, it gets pulled up towards the tail
3. Would the five number summary or mean and standard deviation work better to describe this distribution?  
Five number summary, the mean and st. dev. are affected by outliers and skews in a bad way
4. Does the distribution of lego pieces make sense intuitively? Do you expect there to be more lego sets with smaller number of pieces than larger number of pieces?  
Yes once you remember that there are lego sets (Duplo) designed for infants with few, but relatively large, pieces this makes sense. Without that info I might be slightly suspicious.

### Question 3

1. Below is the histogram of the price on Amazon for the lego set. On the side please indicate if it's symmetric or skewed; if there are outliers; and how many modes it has.

Unimodal, at least one outlier although there those observations \*might\* just be the distributions tail....  
The distribution overall is skewed to the right



2. Does the distribution of lego prices make sense intuitively? Did you expect so many lego sets between 0 and (roughly) \$20? What about the lego set pushing \$200?

I actually expected a price to be more centralized in the \$25-50 range myself. And yeah, some lego sets get to be extremely expensive so the \$200 doesn't surprise me that much (look up the lego death star for example)

3. Would the five number summary or mean and standard deviation work better to describe this distribution?

The five number summary because of the skew and possible outliers

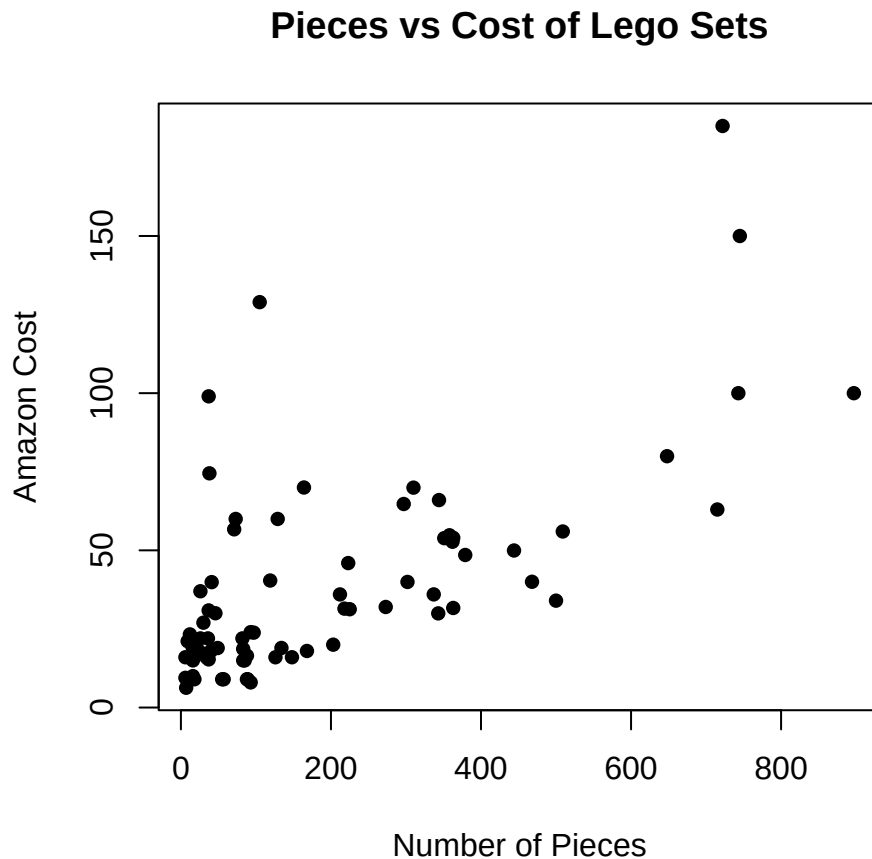
4. Before moving on what would you expect to be the general relationship between the number of pieces and the price? What would you expect to see as the number of pieces goes up?

I'd expect that the number of pieces going up would increase the price simply due to the fact the company is adding more material to the box.

## Question 4: Scatterplots

1. Below is the scatterplot of the price of legos (y-axis) by the number of pieces (x-axis). Please on the side describe the form, direction, outliers, and the strength of the graph.

The graph is roughly positive, weak relationship, and for a lack of a better description I might lean towards linear (although a cloud of points wouldn't be far off from correct). No real outliers that I can tell. Some points are somewhat extreme but they seem to follow the general pattern enough that I wouldn't label them as outliers without more info.



2. Do there seem to be a very useful relationship between the number of pieces and the Amazon cost? Does the number of pieces predict/explain the cost of the lego set on Amazon?

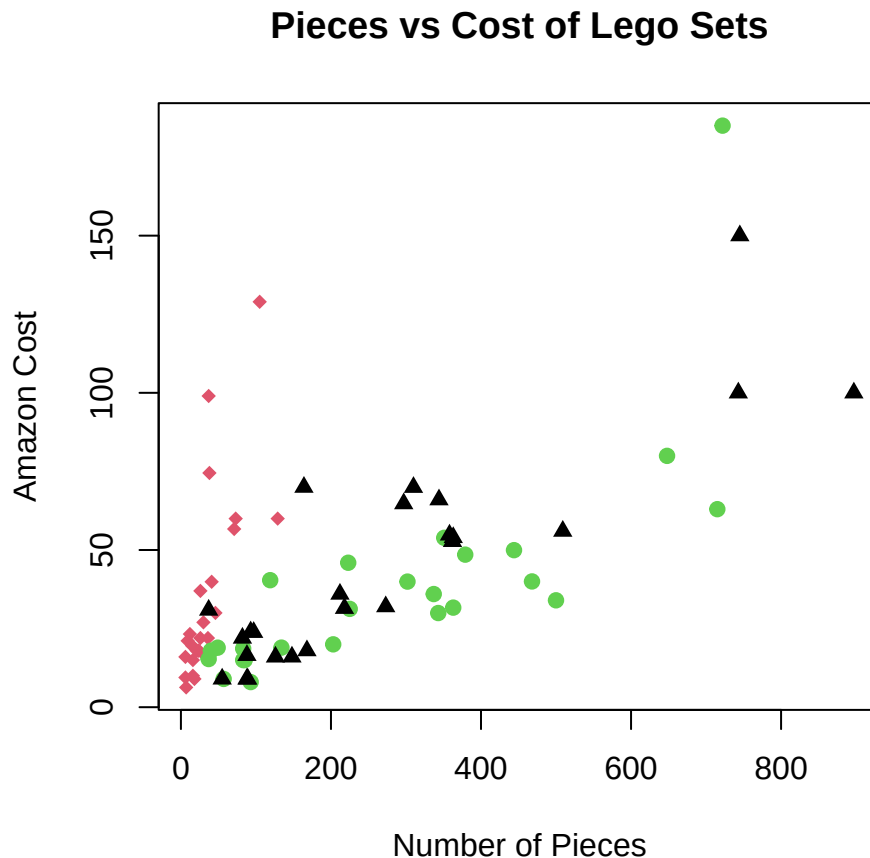
Nope, the relationship looks weak at best with one of the most costly sets having some of the least number of pieces

3. Is the relationship better described by Pearson's or Spearman's correlation coefficient?

Spearman's would probably be best but Pearson's could work to in this situation. The issue is that the relationship is so weak both correlations should return a small, positive value.

## Question 5: Scatterplots by Subpopulations

- Below is the scatterplot of the price of legos (y-axis) by the number of pieces (x-axis). The graph is colored by the Theme. The red diamonds are the DUPLO theme, the green circles are the Friends theme, and the black triangles are the City theme. Please, on the side, mention whether or not the Theme seems to affect the the relationship between the number of pieces and Amazon cost



- Focusing just on the red diamonds (the DUPLO theme), please describe their shape, form, outliers, and direction. Would Pearson's or Spearman's correlation be better?

This is roughly linear with a very steep line up hill (so positive). The strength would be strong or maybe moderate. No meaningful outliers that I see. Pearson's would be best

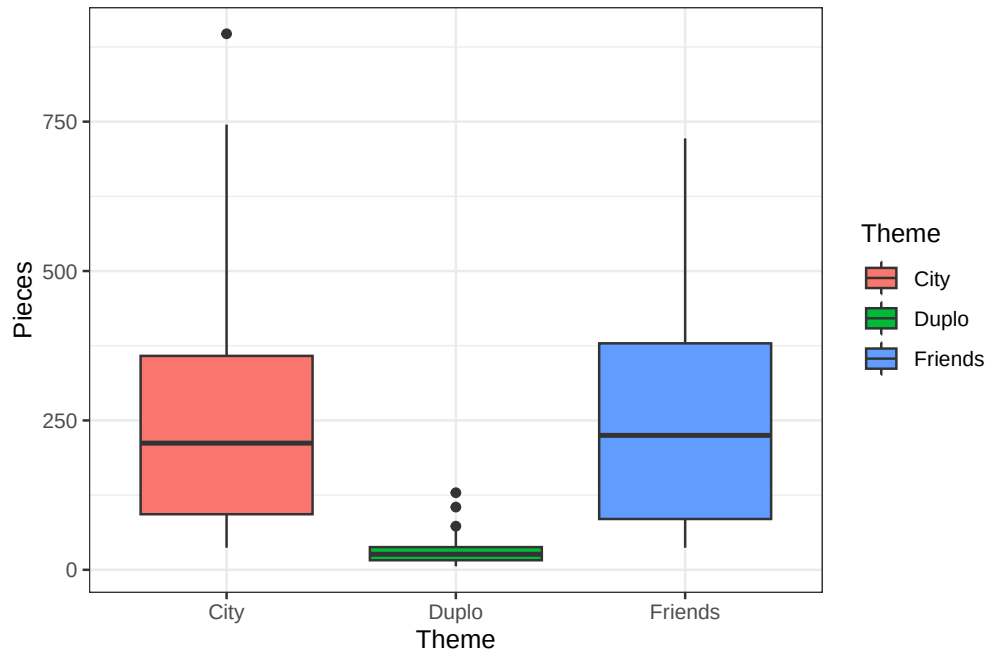
- Focusing just on the green circles (the City theme), please describe their shape, form, outliers, and direction. Would Pearson's or Spearman's correlation be better?

Roughly linear, positive, moderate to strong, and one very arge outlier at Pieces = 700 and Price = \$170. Spearman's would probably be best due to the outlier

## Question 6: Boxplots Revisited

1. Below is the boxplot of the three themes. Please comment on if two themes seem to act similarly. Which one sticks out?

Firends and City act similarly but Duplo is very different



2. Which themes have outliers?

Both city and duplo have outliers

3. Why does Duplo having fewer pieces makes sense?

The Duplo theme is marketed towards infants who can use blocks to develop hand-eye coordination with but with the added requirement the blocks are large to prevent choking. Since infants don't need/want intricate castles/death stars/boats Lego's doesn't sell Duplo sets that let you build those piece intensive sets.

4. (This question has nothing to do with legos, you just need more practice) Please make a boxplot from the following numbers: -4, -2, 0, 1, 1, 2, 3.

5 number summary: min = -4, max = 3, median = 1,  $Q_1 = -1$ ,  $Q_3 = 1.5$  (again, note the quartiles include the median when being calculated so long as the median is in your data set).